

Lab6-assignment-topic-classification

March 16, 2025

0.1 Lab6-Assignment: Topic Classification

Use the same training, development, and test partitions of the the 20 newsgroups text dataset as in Lab6.4-Topic-classification-BERT.ipynb

- Fine-tune and examine the performance of another transformer-based pretrained language models, e.g., RoBERTa, XLNet
- Compare the performance of this model to the results achieved in Lab6.4-Topic-classification-BERT.ipynb and to a conventional machine learning approach (e.g., SVM, Naive Bayes) using bag-of-words or other engineered features of your choice. Describe the differences in performance in terms of Precision, Recall, and F1-score evaluation metrics.

```
[6]: import pandas as pd
import numpy as np
from sklearn.metrics import classification_report
from simpletransformers.classification import ClassificationModel, \
    ClassificationArgs
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_20newsgroups
from collections import Counter
from sklearn.model_selection import train_test_split
```

```
[30]: import re
from wordcloud import WordCloud
import seaborn as sns
from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer
from sklearn.svm import SVC
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
```

Data preprocessing for RoBERTa

```
[7]: categories = ['alt.atheism', 'comp.graphics', 'sci.med', 'sci.space']

training_set = fetch_20newsgroups(subset='train', remove=('headers', 'footers', \
    'quotes'), categories=categories, random_state=42)
testing_set = fetch_20newsgroups(subset='test', remove=('headers', 'footers', \
    'quotes'), categories=categories, random_state=42)
```

```
print(f"Training set: {Counter(training_set.target)}")
print(f"Testing set: {Counter(testing_set.target)}")
```

Training set: Counter({2: 594, 3: 593, 1: 584, 0: 480})
 Testing set: Counter({2: 396, 3: 394, 1: 389, 0: 319})

```
[8]: train_dataframe = pd.DataFrame({'text': training_set.data, 'labels':  
    ↪ training_set.target})  
print(len(train_dataframe))  
train_dataframe.head(5)
```

2251

```
[8]:
```

	text	labels
0	WHile we are on the subject of the shuttle sof...	3
1	There is a program called Graphic Workshop you...	1
2		2
3	My girlfriend is in pain from kidney stones. S...	2
4	I think that's the correct spelling..\n\tI am ...	2

```
[9]: test_dataframe = pd.DataFrame({'text': testing_set.data, 'labels': testing_set.  
    ↪ target})  
print(len(test_dataframe))  
test_dataframe.head(5)
```

1498

```
[9]:
```

	text	labels
0	\nAnd guess who's here in your place.\n\nPleas...	1
1	Does anyone know if any of Currier and Ives et...	1
2	=FLAME ON\n=\n=Reading through the posts about...	2
3	\nBut in this case I said I hoped that BCCI wa...	0
4	\nIn the kind I have made I used a Lite sour c...	2

```
[10]: training, testing = train_test_split(train_dataframe, test_size=0.1,  
    ↪ random_state=0, stratify=train_dataframe[['labels']])  
  
print(f"Training set length: {len(training)}")  
print(f"Testing set length: {len(testing)}")  
print()  
print(f"train: {training[['labels']].value_counts(sort=False)}")  
print(f"test: {testing[['labels']].value_counts(sort=False)}")  
print()  
print(training.head(3))  
print(testing.head(3))
```

Training set length: 2025
 Testing set length: 226

```

train: labels
0      432
1      525
2      534
3      534
Name: count, dtype: int64
test: labels
0      48
1      59
2      60
3      59
Name: count, dtype: int64

```

	text	labels
559	I wonder how many atheists out there care to s...	0
2060	We are interested in purchasing a grayscale pr...	1
1206	Dear Binary Newsers,\n\nI am looking for Quick...	1
	text	labels
1570	I'd dump him. Rude is rude and it seems he en...	2
1761	Hi Everyone ::\n\nI am looking for some soft...	1
455	A friend of mine has been diagnosed with Psori...	2

0.2 Using RoBERTa for topic classification

```

[11]: # Model configuration

roberta_args = ClassificationArgs()

roberta_args.override_output_dir=True
roberta_args.evaluate_during_training=True

roberta_args.num_train_epochs=10
roberta_args.train_batch_size=32
roberta_args.learning_rate=4e-6
roberta_args.max_seq_length=256

roberta_args.use_early_stopping=True
roberta_args.early_stopping_delta=0.01

roberta_args.early_stopping_metric='eval_loss'
roberta_args.early_stopping_metric_minimize=True
roberta_args.early_stopping_patience=2
roberta_args.evaluate_during_training_steps=32

# Kept the same as BERT to equally compare the models

```

```

steps_per_epoch = int(np.ceil(len(training) / float(roberta_args.
    ↪train_batch_size)))
print('Each epoch will have {:,} steps.'.format(steps_per_epoch))

```

Each epoch will have 64 steps.

```

[12]: roberta_model = ClassificationModel('roberta', 'roberta-base', num_labels=4,
    ↪args=roberta_args, use_cuda=False)

```

```
config.json: 0%|          | 0.00/481 [00:00<?, ?B/s]
```

```
model.safetensors: 0%|          | 0.00/499M [00:00<?, ?B/s]
```

Some weights of RobertaForSequenceClassification were not initialized from the model checkpoint at roberta-base and are newly initialized:

```
['classifier.dense.bias', 'classifier.dense.weight', 'classifier.out_proj.bias',
'classifier.out_proj.weight']
```

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```
tokenizer_config.json: 0%|          | 0.00/25.0 [00:00<?, ?B/s]
```

```
vocab.json: 0%|          | 0.00/899k [00:00<?, ?B/s]
```

```
merges.txt: 0%|          | 0.00/456k [00:00<?, ?B/s]
```

```
tokenizer.json: 0%|          | 0.00/1.36M [00:00<?, ?B/s]
```

```

[ ]: #XLNet = ClassificationModel('xlnet', 'xlnet-base-cased', num_labels=4,
    ↪args=roberta_args, use_cuda=False)

```

```

[13]: print(str(roberta_model.args).replace(',', '\n'))

```

```

ClassificationArgs(adafactor_beta1=None
adafactor_clip_threshold=1.0
adafactor_decay_rate=-0.8
adafactor_eps=(1e-30
0.001)
adafactor_relative_step=True
adafactor_scale_parameter=True
adafactor_warmup_init=True
adam_betas=(0.9
0.999)
adam_epsilon=1e-08
best_model_dir='outputs/best_model'
cache_dir='cache_dir/'
config={}
cosine_schedule_num_cycles=0.5
custom_layer_parameters=[]
custom_parameter_groups=[]
dataloader_num_workers=0
do_lower_case=False

```

```
dynamic_quantize=False
early_stopping_consider_epochs=False
early_stopping_delta=0.01
early_stopping_metric='eval_loss'
early_stopping_metric_minimize=True
early_stopping_patience=2
encoding=None
eval_batch_size=100
evaluate_during_training=True
evaluate_during_training_silent=True
evaluate_during_training_steps=32
evaluate_during_training_verbose=False
evaluate_each_epoch=True
fp16=False
gradient_accumulation_steps=1
learning_rate=4e-06
local_rank=-1
logging_steps=50
loss_type=None
loss_args={}
manual_seed=None
max_grad_norm=1.0
max_seq_length=256
model_name='roberta-base'
model_type='roberta'
multiprocessing_chunksize=-1
n_gpu=1
no_cache=False
no_save=False
not_saved_args=[]
num_train_epochs=10
optimizer='AdamW'
output_dir='outputs/'
overwrite_output_dir=True
polynomial_decay_schedule_lr_end=1e-07
polynomial_decay_schedule_power=1.0
process_count=10
quantized_model=False
reprocess_input_data=True
save_best_model=True
save_eval_checkpoints=True
save_model_every_epoch=True
save_optimizer_and_scheduler=True
save_steps=2000
scheduler='linear_schedule_with_warmup'
silent=False
skip_special_tokens=True
tensorboard_dir=None
```

```

thread_count=None
tokenizer_name='roberta-base'
tokenizer_type=None
train_batch_size=32
train_custom_parameters_only=False
trust_remote_code=False
use_cached_eval_features=False
use_early_stopping=True
use_hf_datasets=False
use_multiprocessing=True
use_multiprocessing_for_evaluation=True
wandb_kwargs={}
wandb_project=None
warmup_ratio=0.06
warmup_steps=0
weight_decay=0.0
model_class='ClassificationModel'
labels_list=[0
1
2
3]
labels_map={}
lazy_delimiter='\t'
lazy_labels_column=1
lazy_loading=False
lazy_loading_start_line=1
lazy_text_a_column=None
lazy_text_b_column=None
lazy_text_column=0
onnx=False
regression=False
sliding_window=False
special_tokens_list=[]
stride=0.8
tie_value=1)

```

```
[14]: _, roberta_history = roberta_model.train_model(training, eval_df=testing)
```

```
0%|          | 0/4 [00:00<?, ?it/s]
```

huggingface/tokenizers: The current process just got forked, after parallelism has already been used. Disabling parallelism to avoid deadlocks...

To disable this warning, you can either:

- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true | false)

huggingface/tokenizers: The current process just got forked, after parallelism has already been used. Disabling parallelism to avoid deadlocks...

To disable this warning, you can either:

```

- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)
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has already been used. Disabling parallelism to avoid deadlocks...
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false)
huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)
Epoch:  0%|          | 0/10 [00:00<?, ?it/s]
Running Epoch 1 of 10:  0%|          | 0/64 [00:00<?, ?it/s]
Oit [00:00, ?it/s]
huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)
Oit [00:00, ?it/s]
huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)
Oit [00:00, ?it/s]
huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

```

```

Running Epoch 2 of 10:   0%|               | 0/64 [00:00<?, ?it/s]
Oit [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
    - Avoid using `tokenizers` before the fork if possible
    - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)
Oit [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
    - Avoid using `tokenizers` before the fork if possible
    - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)
Oit [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
    - Avoid using `tokenizers` before the fork if possible
    - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)
Oit [00:00, ?it/s]

Running Epoch 3 of 10:   0%|               | 0/64 [00:00<?, ?it/s]
Oit [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
    - Avoid using `tokenizers` before the fork if possible
    - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)
Oit [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
    - Avoid using `tokenizers` before the fork if possible
    - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)
Oit [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
    - Avoid using `tokenizers` before the fork if possible
    - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)
Oit [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
    - Avoid using `tokenizers` before the fork if possible
    - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)
Oit [00:00, ?it/s]

```


- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true | false)

Running Epoch 4 of 10: 0%| | 0/64 [00:00<?, ?it/s]

Oit [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism has already been used. Disabling parallelism to avoid deadlocks...

To disable this warning, you can either:

- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true | false)

Oit [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism has already been used. Disabling parallelism to avoid deadlocks...

To disable this warning, you can either:

- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true | false)

Oit [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism has already been used. Disabling parallelism to avoid deadlocks...

To disable this warning, you can either:

- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true | false)

Running Epoch 5 of 10: 0%| | 0/64 [00:00<?, ?it/s]

Oit [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism has already been used. Disabling parallelism to avoid deadlocks...

To disable this warning, you can either:

- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true | false)

Oit [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism has already been used. Disabling parallelism to avoid deadlocks...

To disable this warning, you can either:

- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true | false)

Oit [00:00, ?it/s]

```

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
    - Avoid using `tokenizers` before the fork if possible
    - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

Running Epoch 6 of 10:   0%|               | 0/64 [00:00<?, ?it/s]

Oit [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
    - Avoid using `tokenizers` before the fork if possible
    - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

Oit [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
    - Avoid using `tokenizers` before the fork if possible
    - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

Oit [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
    - Avoid using `tokenizers` before the fork if possible
    - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

Running Epoch 7 of 10:   0%|               | 0/64 [00:00<?, ?it/s]

Oit [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
    - Avoid using `tokenizers` before the fork if possible
    - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

Oit [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
    - Avoid using `tokenizers` before the fork if possible

```

```

- Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)
Oit [00:00, ?it/s]
huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)
Running Epoch 8 of 10:  0%|          | 0/64 [00:00<?, ?it/s]
Oit [00:00, ?it/s]
huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

```

0.3 Model evaluation

```

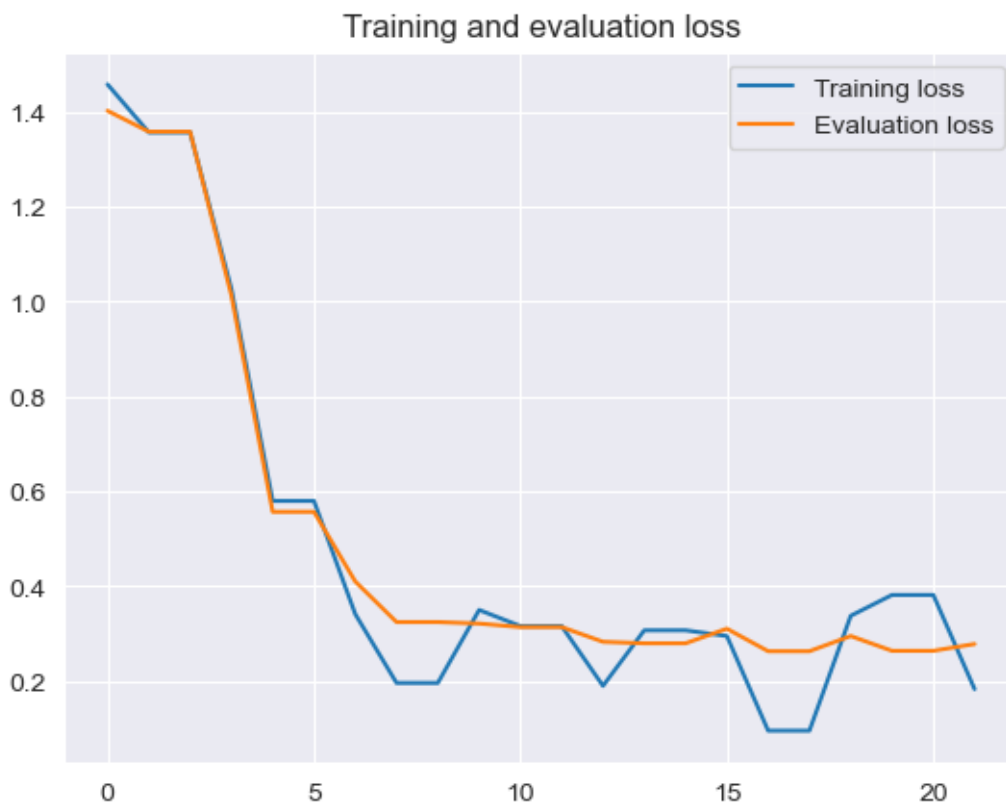
[15]: # Training and evaluation loss
train_loss = roberta_history['train_loss']
eval_loss = roberta_history['eval_loss']
plt.plot(train_loss, label='Training loss')
plt.plot(eval_loss, label='Evaluation loss')
plt.title('Training and evaluation loss')
plt.legend()

```

```

[15]: <matplotlib.legend.Legend at 0x30d1db980>

```



```
[25]: result, model_outputs, wrong_predictions = roberta_model.eval_model(testing)
      result
```

```
0it [00:00, ?it/s]
```

huggingface/tokenizers: The current process just got forked, after parallelism has already been used. Disabling parallelism to avoid deadlocks...

To disable this warning, you can either:

- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true | false)

```
Running Evaluation:  0%|          | 0/3 [00:00<?, ?it/s]
```

```
[25]: {'mcc': 0.8475537969229284, 'eval_loss': 0.2784139464298884}
```

```
[26]: predicted, probabilities = roberta_model.predict(test_dataframe.text.to_list())
      test_dataframe['predicted'] = predicted
```

```
0%|          | 0/2 [00:00<?, ?it/s]
```

huggingface/tokenizers: The current process just got forked, after parallelism has already been used. Disabling parallelism to avoid deadlocks...

To disable this warning, you can either:

```

- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)
huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)
huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

0%|          | 0/15 [00:00<?, ?it/s]

```

```
[27]: test_dataframe.head(5)
```

```
[27]:
```

	text	labels	predicted
0	\nAnd guess who's here in your place.\n\nPleas...	1	1
1	Does anyone know if any of Currier and Ives et...	1	1
2	=FLAME ON\n=\n=Reading through the posts about...	2	2
3	\nBut in this case I said I hoped that BCCI wa...	0	0
4	\nIn the kind I have made I used a Lite sour c...	2	2

```
[28]: print(classification_report(test_dataframe['labels'],
↳test_dataframe['predicted']))
```

	precision	recall	f1-score	support
0	0.85	0.78	0.81	319
1	0.90	0.90	0.90	389
2	0.84	0.92	0.88	396
3	0.83	0.81	0.82	394
accuracy			0.86	1498
macro avg	0.86	0.85	0.85	1498
weighted avg	0.86	0.86	0.86	1498

Data preprocessing for SVM

```
[40]: data_train, labels_train = training_set.data, training_set.target
data_test, labels_test = testing_set.data, testing_set.target

vectorizer = TfidfVectorizer(max_features=999999999999999, stop_words='english')
data_train_tfidf = vectorizer.fit_transform(data_train)
```

```
data_test_tfidf = vectorizer.transform(data_test)

print(data_train_tfidf.shape)
print('-----')
print(data_test_tfidf.shape)
```

```
(2251, 29764)
```

```
-----
(1498, 29764)
```

0.4 Using SVM with tfidf features for topic classification

```
[51]: svm_model = SVC(kernel='linear', C=1.0)
      svm_model.fit(data_train_tfidf, labels_train)
```

```
[51]: SVC(kernel='linear')
```

```
[52]: y_pred = svm_model.predict(data_test_tfidf)
```

```
[53]: print(classification_report(labels_test, y_pred, target_names=categories))
```

	precision	recall	f1-score	support
alt.atheism	0.85	0.75	0.80	319
comp.graphics	0.88	0.87	0.88	389
sci.med	0.87	0.83	0.85	396
sci.space	0.75	0.87	0.81	394
accuracy			0.83	1498
macro avg	0.84	0.83	0.83	1498
weighted avg	0.84	0.83	0.83	1498

0.5 Comparison of results

The BERT-RoBERTa model scores a weighted average score of 0.86 for its precision, recall and f1-score, and the SVM scores a weighted average of 0.84 for precision and a score of 0.83 for recall and f1-score. This shows that on average BERT-RoBERTa outperforms SVM, thus on average, BERT-RoBERTa is more accurate and precise. Looking more specifically at the results, for classifying as atheism, BERT and SVM score the same precision score, but BERT has a better recall score of 0.78 compared to 0.75. For the graphics topic, BERT had a higher precision and recall score. For the topic of medicine, BERT scored 0.84 for precision, and SVM scored 0.87 here. However, this is contrasted by their recall scores of 0.92 and 0.83, respectively; the higher F1 score for BERT shows that the higher recall score was enough to compensate for this. A similar but reversed instance is present when assessing the scores for the space topic. Here, BERT underperforms based on the Recall score, however, based on the F1-score compensates for this with a higher score for precision. In conclusion, it is clear that the BERT-RoBERTa model performed better than the SVM, with SVM sparingly, but inconsequentially performing better. However, the difference between the weighted average of BERT, 0.86, and SVM, 0.83, is small and whilst BERT

performed better with this dataset, this does not have to be correct in every case. Furthermore, SVM and BERT could be fine-tuned and thus score higher. Further research could investigate the results of different and more difficult datasets.